

```
MariaDB [(none)]> create database Assignment_2;
Query OK, 1 row affected (0.046 sec)
```

```
MariaDB [(none)]> use Assignment_2;
Database changed
```

```
MariaDB [Assignment_2]> create table Employee(
-> empNo int not null primary key,
-> fName varchar(20),
-> lName varchar(20),
-> address varchar(25),
-> DOB date,
-> sex char(1) not null,
-> constraint check_sex check(sex in ('M','F')),
-> position varchar(50) check (position in ('Manager', 'Team Leader',
'Analyst', 'Software Developer')),
-> hoursWorked int check(hoursWorked >= 0 and hoursWorked <= 40));
Query OK, 0 rows affected (0.099 sec)
```

```
MariaDB [Assignment_2]> alter table Employee add deptNo int;
Query OK, 0 rows affected (0.053 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [Assignment_2]> desc Employee;
```

Field	Type	Null	Key	Default	Extra
empNo	int(11)	NO	PRI	NULL	
fName	varchar(20)	YES		NULL	
lName	varchar(20)	YES		NULL	
address	varchar(25)	YES		NULL	
DOB	date	YES		NULL	
sex	char(1)	NO		NULL	
position	varchar(50)	YES		NULL	
hoursWorked	int(11)	YES		NULL	
deptNo	int(11)	YES		NULL	

```
9 rows in set (0.050 sec)
```

```
MariaDB [Assignment_2]> create table Department (
-> deptNo int not null primary key,
-> deptName varchar(50),
-> mgEmpNo int);
Query OK, 0 rows affected (0.007 sec)
```

```
MariaDB [Assignment_2]> describe Department;
```

Field	Type	Null	Key	Default	Extra
deptNo	int(11)	NO	PRI	NULL	
deptName	varchar(50)	YES		NULL	
mgEmpNo	int(11)	YES		NULL	

```
3 rows in set (0.031 sec)
```

```
MariaDB [Assignment_2]> create table Project(
-> projNo int not null primary key,
-> projName varchar(50),
-> deptNo int);
Query OK, 0 rows affected (0.033 sec)
```

```
MariaDB [Assignment_2]> desc Project;
```

Field	Type	Null	Key	Default	Extra
projNo	int(11)	NO	PRI	NULL	
projName	varchar(50)	YES		NULL	
deptNo	int(11)	YES		NULL	

projNo	int(11)	NO	PRI	NULL		
projName	varchar(50)	YES		NULL		
deptNo	int(11)	YES		NULL		

3 rows in set (0.032 sec)

```
MariaDB [Assignment_2]> create table WorksOn(
-> empNo int,
-> projNo int,
-> dateWorked date,
-> hoursWorked decimal(5,2),
-> primary key (empNo,projNo,dateWorked),
-> foreign key (empNo) references Employee(empNo),
-> foreign key (projNo) references Project(projNo));
Query OK, 0 rows affected (0.039 sec)
```

```
MariaDB [Assignment_2]> desc WorksOn;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| empNo | int(11) | NO | PRI | NULL | |
| projNo | int(11) | NO | PRI | NULL | |
| dateWorked | date | NO | PRI | NULL | |
| hoursWorked | decimal(5,2) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.032 sec)
```

```
MariaDB [Assignment_2]> alter table department add constraint foreign key
(mgrEmpNo) references Employee(empNo);
ERROR 1072 (42000): Key column 'mgrEmpNo' doesn't exist in table
MariaDB [Assignment_2]> alter table department add constraint foreign key
(mgEmpNo) references Employee(empNo);
Query OK, 0 rows affected (0.049 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [Assignment_2]> desc Department
-> ;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| deptNo | int(11) | NO | PRI | NULL | |
| deptName | varchar(50) | YES | | NULL | |
| mgEmpNo | int(11) | YES | MUL | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.029 sec)
```

```
MariaDB [Assignment_2]> alter table Project add constraint foreign key (deptNo)
references Department(deptNo);
Query OK, 0 rows affected (0.042 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

POPULATING TABLES

```
a)
MariaDB [assignment_2]> INSERT INTO department (deptNo, deptName, mgEmpNo)
VALUES
-> (10, 'Management', NULL),
-> (20, 'Research & Development', NULL),
-> (30, 'IT Support', NULL),
-> (40, 'Cloud Infrastructure', NULL),
-> (50, 'Data Analytics', NULL),
-> (60, 'Quality Assurance', NULL),
-> (70, 'Cybersecurity', NULL),
-> (80, 'Product Design', NULL),
-> (90, 'DevOps', NULL),
```

```
-> (100, 'Human Resources', NULL);
Query OK, 10 rows affected (0.026 sec)
```

```
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1001 WHERE deptNo = 10;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1002 WHERE deptNo = 20;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1010 WHERE deptNo = 30;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1011 WHERE deptNo = 40;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1012 WHERE deptNo = 50;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1013 WHERE deptNo = 60;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1014 WHERE deptNo = 70;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1015 WHERE deptNo = 80;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1016 WHERE deptNo = 90;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1017 WHERE deptNo = 100;
```

```
MariaDB [assignment_2]> SELECT * FROM department;
```

deptNo	deptName	mgEmpNo
10	Management	1001
20	Research & Development	1002
30	IT Support	1010
40	Cloud Infrastructure	1011
50	Data Analytics	1012
60	Quality Assurance	1013
70	Cybersecurity	1014
80	Product Design	1015
90	DevOps	1016
100	Human Resources	1017

```
10 rows in set (0.001 sec)
```

b)

```
MariaDB [assignment_2]> INSERT INTO project (projNo, projName, deptNo) VALUES
```

```
-> (501, 'Enterprise ERP', 10),
-> (502, 'NextGen AI Engine', 20),
-> (503, 'Cloud Migration', 40),
-> (504, 'Big Data Warehouse', 50),
-> (505, 'Security Audit 2026', 70),
-> (506, 'UI/UX Redesign', 80),
-> (507, 'CI/CD Automation', 90),
-> (508, 'Mobile App v2', 20),
-> (509, 'Helpdesk Overhaul', 30),
-> (510, 'Predictive Analytics', 50);
```

```
Query OK, 10 rows affected (0.025 sec)
```

```
Records: 10 Duplicates: 0 Warnings: 0
```

```
MariaDB [assignment_2]> select * from project;
```

projNo	projName	deptNo
501	Enterprise ERP	10
502	NextGen AI Engine	20
503	Cloud Migration	40
504	Big Data Warehouse	50
505	Security Audit 2026	70
506	UI/UX Redesign	80
507	CI/CD Automation	90
508	Mobile App v2	20
509	Helpdesk Overhaul	30
510	Predictive Analytics	50

```
10 rows in set (0.001 sec)
```

c)

```
MariaDB [assignment_2]> INSERT INTO workson (empNo, projNo, dateWorked, hoursWorked) VALUES
```

```
-> (1001, 501, '2026-06-01', 10.00),
-> (1003, 502, '2026-06-01', 40.00),
-> (1004, 502, '2026-06-01', 35.50),
-> (1004, 502, '2026-06-02', 25.00),
-> (1005, 504, '2026-06-02', 38.00),
-> (1006, 503, '2026-06-03', 40.00),
-> (1007, 503, '2026-06-03', 20.25),
-> (1008, 504, '2026-06-04', 30.00),
-> (1009, 502, '2026-06-04', 15.00),
-> (1009, 510, '2026-06-05', 12.50);
```

Query OK, 10 rows affected (0.002 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
MariaDB [assignment_2]> select * from workson;
```

empNo	projNo	dateWorked	hoursWorked
1001	501	2026-06-01	10.00
1003	502	2026-06-01	40.00
1004	502	2026-06-01	35.50
1004	502	2026-06-02	25.00
1005	504	2026-06-02	38.00
1006	503	2026-06-03	40.00
1007	503	2026-06-03	20.25
1008	504	2026-06-04	30.00
1009	502	2026-06-04	15.00
1009	510	2026-06-05	12.50

10 rows in set (0.001 sec)

d)

```
INSERT INTO employee (empNo, fName, lName, address, DOB, sex, position, hoursWorked, deptNo) VALUES
```

```
-> (1001, 'James', 'Smith', '123 Pine St, London', '1980-05-12', 'M', 'Manager', 40, 10),
-> (1002, 'Sarah', 'Jones', '456 Oak Ave, Manchester', '1985-08-24', 'F', 'Manager', 40, 20),
-> (1003, 'Michael', 'Brown', '789 Maple Rd, Bristol', '1988-11-02', 'M', 'Team Leader', 38, 20),
-> (1004, 'Emily', 'Davis', '101 Cedar Ln, Leeds', '1992-03-15', 'F', 'Software Developer', 35, 20),
-> (1005, 'David', 'Wilson', '202 Birch Dr, Liverpool', '1990-07-19', 'M', 'Analyst', 40, 50),
-> (1006, 'Jessica', 'Taylor', '303 Ash Mews, Newcastle', '1993-01-30', 'F', 'Software Developer', 40, 40),
-> (1007, 'Ashley', 'Thomas', '404 Elm Blvd, Sheffield', '1987-09-10', 'M', 'Team Leader', 37, 40),
-> (1008, 'Matthew', 'White', '505 Willow Way, Bham', '1991-06-05', 'M', 'Analyst', 20, 50),
-> (1009, 'Amanda', 'Martin', '606 Cherry Ct, Notts', '1994-12-14', 'F', 'Software Developer', 30, 20),
-> (1010, 'Robert', 'Clark', '707 Plum La, Southampton', '1983-04-22', 'M', 'Manager', 40, 30);
```

```
MariaDB [assignment_2]> select * from employee;
```

empNo	fName	lName	address	DOB	sex	position	hoursWorked	deptNo
-------	-------	-------	---------	-----	-----	----------	-------------	--------

```

+-----+-----+-----+
| 1001 | James | Smith | 123 Pine St, London | 1980-05-12 | M |
Manager | 40 | 10 |
| 1002 | Sarah | Jones | 456 Oak Ave, Manchester | 1985-08-24 | F |
Manager | 40 | 20 |
| 1003 | Michael | Brown | 789 Maple Rd, Bristol | 1988-11-02 | M | Team
Leader | 38 | 20 |
| 1004 | Emily | Davis | 101 Cedar Ln, Leeds | 1992-03-15 | F |
Software Developer | 35 | 20 |
| 1005 | David | Wilson | 202 Birch Dr, Liverpool | 1990-07-19 | M |
Analyst | 40 | 50 |
| 1006 | Jessica | Taylor | 303 Ash Mews, Newcastle | 1993-01-30 | F |
Software Developer | 40 | 40 |
| 1007 | Ashley | Thomas | 404 Elm Blvd, Sheffield | 1987-09-10 | M | Team
Leader | 37 | 40 |
| 1008 | Matthew | White | 505 Willow Way, Bham | 1991-06-05 | M |
Analyst | 20 | 50 |
| 1009 | Amanda | Martin | 606 Cherry Ct, Notts | 1994-12-14 | F |
Software Developer | 30 | 20 |
| 1010 | Robert | Clark | 707 Plum La, Southampton | 1983-04-22 | M |
Manager | 40 | 30 |
+-----+-----+-----+

```

10 rows in set (0.002 sec)

2)

```

MariaDB [assignment_2]> create view ProjManagedbyFemale as select
project.projNo,project.projName,project.deptNo,department.deptName,employee.fNam
e as ManagerFirstName,employee.lName as ManagerLastName from project inner join
department on project.deptNo = department.deptNo inner join employee on
department.mgEmpNo = employee.empNo where employee.sex = 'F' order by
project.projNo;
Query OK, 0 rows affected (0.041 sec)

```

```

MariaDB [assignment_2]> select * from ProjManagedbyFemale;
+-----+-----+-----+-----+
| projNo | projName | deptNo | deptName |
ManagerFirstName | ManagerLastName |
+-----+-----+-----+-----+
| 502 | NextGen AI Engine | 20 | Research & Development | Sarah
Jones |
| 508 | Mobile App v2 | 20 | Research & Development | Sarah
Jones |
+-----+-----+-----+-----+
2 rows in set (0.015 sec)

```

3)

a)

```

MariaDB [assignment_2]> CREATE VIEW EmpProject (empNo, projNo, totalHours) AS
-> SELECT w.empNo, w.projNo, SUM(w.hoursWorked)
-> FROM employee e, project p, workson w
-> WHERE e.empNo = w.empNo AND p.projNo = w.projNo
-> GROUP BY w.empNo, w.projNo;
Query OK, 0 rows affected (0.007 sec)

```

```

MariaDB [assignment_2]> select * from EmpProject;
+-----+-----+-----+
| empNo | projNo | totalHours |
+-----+-----+-----+
| 1001 | 501 | 10.00 |
| 1003 | 502 | 40.00 |

```

1004	502	60.50
1005	504	38.00
1006	503	40.00
1007	503	20.25
1008	504	30.00
1009	502	15.00
1009	510	12.50

9 rows in set (0.006 sec)

b)

```
MariaDB [assignment_2]> select projNo from EmpProject where projNo = 502;
```

projNo
502
502
502

3 rows in set (0.002 sec)

c)

```
MariaDB [assignment_2]> select count(projNo) from EmpProject where empNo = 1001;
```

count(projNo)
1

1 row in set (0.001 sec)

d)

```
MariaDB [assignment_2]> select empNo,totalHours from EmpProject group by empNo;
```

empNo	totalHours
1001	10.00
1003	40.00
1004	60.50
1005	38.00
1006	40.00
1007	20.25
1008	30.00
1009	15.00

8 rows in set (0.005 sec)